

# Muting (Don) Ma, Ph.D.

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 Operations Management Ph.D. Program - University of Alabama

*Last Updated: February 23, 2026*

## Operations Research Scholar · Optimization Theorist · Quantitative Methodologist

Operations researcher with dual Ph.D. training and **8+** years of scholarly experience, specializing in **optimization theory, game-theoretic modeling, and market dynamics**. Published in leading journals including *Pattern Recognition* and *Transportation Research Part C*. Lead contributor on **\$500K+** in competitively funded research. Committed to advancing theoretical foundations while mentoring the next generation of scholars.

## Education

- 2025–Present**    **Ph.D. in Operations Management** (In Progress), University of Alabama  
*Culverhouse College of Business, Department of Information Systems, Statistics, and Management Science (ISM)*  
*Research Focus: Competitive dynamics modeling, Nash equilibrium analysis, sustainable operations*  
*Advisor: Dr. Mesut Yavuz*  
*Award: 2nd Place, Culverhouse Data Analytics Summit Poster Competition (2025)*
- 2017–2022**    **Ph.D. in Civil Engineering**, University of Louisville  
*Department of Civil and Environmental Engineering*  
*Dissertation: Optimal scheduling of connected and autonomous vehicles at reservation-based intersections*  
*Advisor: Dr. Zhixia Li*  
*Awards: Grosscurth Fellowship (2017), Doctoral Dissertation Completion Award (2022)*

## Peer-Reviewed Publications

- 2025**    **Ma, M., Yavuz, M., et al.** “C2SLM: A Correlation-based Clustering-assisted Sparse Learning Model for Electric Vehicle Market Demand Forecasting.” *Pattern Recognition*, 170, p.111984. DOI: [10.1016/j.patcog.2025.111984](https://doi.org/10.1016/j.patcog.2025.111984)
- Correlation-polarized feature generation with binary dummy and interaction terms
  - Agglomerative hierarchical clustering with panel-based unobserved heterogeneity
  - Clustering-assisted Lasso (caLasso) for sparse estimation on econometric panel data
- 2023**    **Ma, M. & Li, Z.** “A speed-maximization trajectory optimization model on a reservation based intersection control system.” *Transportation Research Part C: Emerging Technologies*, 154, p.104266. DOI: [10.1016/j.trc.2023.104266](https://doi.org/10.1016/j.trc.2023.104266)
- Discrete-time mixed integer programming with queue-theoretic objective derivation
  - Little’s Law–based analytical proof of speed-maximization optimality
  - Piecewise continuous-time trajectory modeling with provable feasibility guarantees
- 2021**    **Ma, M. & Li, Z.** “A time-independent trajectory optimization approach for connected and autonomous vehicles.” *Transportation Research Interdisciplinary Perspectives*, 9, p.100312. DOI: [10.1016/j.trip.2021.100312](https://doi.org/10.1016/j.trip.2021.100312)
- Integer linear programming with variable acceleration formulation
  - Cell-based reservation scheduling with enlarged trajectory segmentation
  - BATCH and ZONE control strategies for high-demand adaptation

## Working Papers & Papers Under Review

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- Under Review**    **Ma, M.,** Li, Z., & Yavuz, M. “Reservation-based Intersection Scheduling using Dynamic Programming with Dominance Pruning.” *International Transactions in Operational Research*. [SSRN: 4097173](#)
- *Dynamic programming with dominance pruning (DP<sup>2</sup>) and analytical lower-bound node elimination*
  - *$O(n^2)$  worst-case and  $\Omega(n)$  best-case time complexity*
  - *91%/99% computation speedup over state-of-the-art DP and Gurobi solver*
- Under Review**    **Ma, M.,** Li, Z., Hudnall, M., Wang, Q., & Yavuz, M. “Socioeconomic Tradeoff on Connected and Automated Vehicles Scheduling at a Reservation-based Intersection.” *European Journal of Operational Research*. [SSRN: 5372729](#)
- *Bi-objective optimization with Pareto frontier analysis*
  - *Simulation-optimization-in-the-loop (SOIL) framework*
  - *Fixed-effects and pooled OLS regression; queueing-theoretic delay–collision cost tradeoff*
- Working Paper**    **Ma, M.** “Cognitive Workload Allocation Under Uncertainty: A Stochastic Multiple Knapsack Approach with Workload-Dependent Returns.” [DOI: 10.5281/zenodo.18039750](#)
- *Stochastic multiple knapsack formulation with chance constraints*
  - *Workload-dependent return functions with mean-CVaR risk-averse objectives*
  - *Benders decomposition for scalable solution*

## Technical Reports

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- 2022**                    Li, Z. & **Ma, M.** “A Real-Time Proactive Intersection Safety Monitoring System Based on Radar Sensor Data.” *NCHRP IDEA Project 217*. [Transportation Research Board](#)
- *24GHz Microwave Doppler radar multi-sensor fusion with coordinate transformation*
  - *Automated TTC computation, trajectory noise removal, and movement classification*
  - *Real-time conflict detection: 4.8% severity error rate, 80% true positive rate*

## Research Experience

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- 2025–Present**        **Doctoral Researcher**, University of Alabama  
*Advisor: Dr. Mesut Yavuz, Department of ISM*
- Developing primal-dual optimization frameworks for analyzing manufacturer competition strategies in technology transitions
  - Investigating Nash equilibrium conditions and welfare implications in electric vehicle market adoption
  - Theoretical contributions: Characterizing trade-offs between firm profitability and societal sustainability objectives
- 2022–2025**            **Postdoctoral Research Associate**, University of Alabama  
*Institute of Data & Analytics, Culverhouse College of Business*  
*Research Areas: Machine learning, bi-objective optimization, large-scale simulation*
- Developed C2SLM framework—novel sparse learning methodology combining correlation-based clustering with feature selection
  - Designed system dynamics models for technology adoption forecasting with theoretical grounding in diffusion theory
  - Led interdisciplinary research team of 8+ members; mentored undergraduate, master’s, and doctoral students
  - Lead researcher role: Alabama Mobility and Power Center (**\$287,751**)
- 2017–2022**            **Doctoral Researcher**, University of Louisville  
*Advisor: Dr. Zhixia Li, Department of Civil and Environmental Engineering*
- Developed theoretical foundations for optimal vehicle scheduling at signal-free intersections using dynamic programming
  - Formulated novel trajectory optimization models with provable optimality conditions
  - Extended dominance pruning techniques to reduce computational complexity in large-scale scheduling problems

- NSF I/UCRC funded research (**\$45,990**)

**2019–2021**

**Graduate Research Assistant**, University of Louisville

*NCHRP IDEA Project 217 (\$137,000) & KYTC Highway Safety Improvement Program (\$39,318)*

- Developed real-time proactive safety monitoring systems with methodological contributions in sensor fusion
- Research outcomes adopted by transportation agencies for practical implementation

## Teaching Experience

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**2025–Present**

**Graduate Teaching Assistant**, University of Alabama

- OM 310: Introduction to Management Science (Undergraduate)  
*Linear programming, decision analysis, and optimization fundamentals*
- OM 524: Manufacturing Scheduling & Control Systems (Graduate)  
*Scheduling algorithms, production planning, and control theory*
- OM 525: Effective Quality Management (Graduate)  
*Statistical quality control, process improvement methodologies*

**2022–2025**

**Research Mentor**, University of Alabama

- Supervised undergraduate research assistants on data analysis projects
- Mentored master's and doctoral students on research methodology and academic writing

## Research Grants & Funding

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**2022–2025**

**Alabama Mobility and Power Center**, \$287,751

*Role: Lead Researcher. Electric vehicle market analysis and demand forecasting.*

**2019–2022**

**NCHRP IDEA Project 217**, \$137,000

*Role: Graduate Research Assistant. Real-time intersection safety monitoring systems.*

**2019–2021**

**KYTC Highway Safety Improvement Program**, \$39,318

*Role: Graduate Research Assistant. Safety countermeasures design and implementation.*

**2017–2022**

**NSF Industry-University Cooperative Research Centers**, \$45,990

*Role: Graduate Research Assistant. Connected vehicle intersection control.*

**Total Funding: \$509,059**

## Advanced Coursework

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**Optimization Theory**

Linear Programming (OM 600), Stochastic Decision Models (OM 601), Integer Modeling and Optimization (OM 603), Foundations of Optimization I (IE 610), Inventory Management (OM 523)

**Statistical Methods**

Mathematical Statistics I (ST 554), Special Topics in Analytics (ST 597), Predictive Modeling (BAN 501), Advanced Mathematical Statistics (ST 697)

**Game Theory**

Game Theory (EC 660)

**Computational Methods**

Data Mining (CECS 632), Transportation Engineering (CEE 560)

## Technical Skills

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**Programming**

C++, C#, Python, Java, SQL

**Optimization Solvers**

Gurobi, CPLEX

**Statistical Software**

R, Tableau

**Development Tools**

Visual Studio, Linux

**Methodological Expertise**

◦ **Optimization**

Linear/Integer Programming, Dynamic Programming, Multi-objective Optimization, Dominance Pruning, Benders Decomposition

- **Stochastic Methods** Stochastic Programming, Chance-constrained Optimization, Queue Theory, Simulation-Optimization
- **Statistical Learning** Sparse Learning (Lasso/Group Lasso), Panel Data Econometrics, Fixed/Random Effects Models, Regression Analysis
- **Game Theory** Nash Equilibrium Analysis, Mechanism Design, Coopetition Modeling
- **Artificial Intelligence** Attention/Transformer Architectures, Gradient Descent Optimization, XGBoost, Neural Network, Clustering, Dimension Reduction, Predictive Modeling

## Awards & Honors

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**2025** 2nd Place, Culverhouse Data Analytics Summit Poster Competition, University of Alabama  
**2022** Doctoral Dissertation Completion Award, University of Louisville  
**2017** Grosscurth Fellowship, University of Louisville

## Professional Service & Affiliations

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**Representative Journal Reviewer** International Transactions on Operational Research, PLOS One, IEEE Transactions on Intelligent Transportation Systems, Energy Systems  
**Memberships** INFORMS, Transportation Research Board